



Numerical Methods · Mark Scheme

Chapter 9: Numerical Methods. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Change of sign

- 1 $f(1) = -1, f(2) = 7$ M1; sign change and f continuous, so root in $(1, 2)$ A1.
- 2 $f(1.5) = -0.0183, f(1.6) = 0.153$ M1 A1; sign change, f continuous, so root in $[1.5, 1.6]$ A1.
- 3 Interval $[2.125, 2.135]$ M1; $f(2.125) = -0.0293, f(2.135) = 0.0568$ A1; sign change, continuous, so $\alpha = 2.13$ (2 d.p.) A1.
- 4 $\tan x$ is not continuous on $[1, 2]$ B1; it has an asymptote at $x = \frac{\pi}{2} \approx 1.571$ B1; the sign change is caused by the discontinuity, not a root B1.
- 5(a) $f(2) = 0.386, f(3) = -0.391$ M1; sign change, continuous for $x > -2$ A1.
- 5(b) Interval $[2.505, 2.515]$ M1; $f(2.505) = 0.000188, f(2.515) = -0.00759$ A1; conclusion A1.
- 5(c) $f(-1.99) = -1.615, f(-1.9) = 0.597$ M1; sign change, so second root A1.

Section B – Fixed-point iteration

- 6 $x_2 = 2.2361, x_3 = 2.7335, x_4 = 2.9098$ M1 A1.
- 7 (a) Divide by x^2 : $x - 4 - \frac{1}{x^2} = 0$ B1. (b) 4.0625, 4.0606, 4.0607... M1; root 4.0606 A1.
- 8 $5x = x^2 + 2$ B1; 0.45, 0.4405, 0.4388, 0.4385 M1; root 0.438 A1.
- 9 $e^x = 4 - x \Rightarrow x = \ln(4 - x)$ B1; 1.0986, 1.0652, 1.0766, 1.0727, 1.0741, ... M1 A1; root 1.074 A1.
- 10(a) $f(1.8) = -0.768, f(1.9) = 0.559$ M1; sign change, continuous A1.
- 10(b) $x^3 = 12 - 3x$ M1; cube root A1.
- 10(c) $x_2 = 1.8171$ M1 A1; $x_3 = 1.8709, x_4 = 1.8554$ A1.
- 10(d) $f(1.8585) = -0.00521 < 0, f(1.8595) = 0.00817 > 0$ M1; sign change, so $\alpha = 1.859$ A1.

Section C – Convergence

- 11 $|g'(L)| < 1$, so converges B1; negative, so cobweb (oscillating) B1.
- 12 $g'(x) = \frac{2x}{5}$ B1; $g'(0.438) = 0.175$ M1; $|g'(L)| < 1$ so converges; positive so staircase A1.
- 13 $g'(x) = \frac{3}{4}x^2$ B1; $g'(2.115) = 3.35$ M1; $|g'(\beta)| > 1$, so the iteration diverges from β A1.
- 14 (a) Cobweb spiralling inwards B1; $x_2 = 0.980$ and $x_3 = 0.557$ marked B1. (b) $g'(L) = -\sin 0.739 = -0.674$ M1; $-1 < g'(L) < 0$: convergent cobweb, consistent with diagram A1.
- 15(a) $f(2) = -3, f(3) = 13$ M1; sign change, continuous A1.
- 15(b) A: $3x = x^3 - 5$ B1; B: $x^3 = 3x + 5$ B1.
- 15(c) A: $g'(x) = x^2, g'(2.3) = 5.29 > 1$, diverges M1 A1; B: $g'(x) = (3x + 5)^{-\frac{2}{3}}, g'(2.3) = 0.19 < 1$, converges M1 A1.
- 15(d) 2.2240, 2.2684, 2.2770, 2.2786, ... M1; $\alpha = 2.28$ A1.

Section D – Newton-Raphson

- 16 $f'(x) = 2x$ B1; $x_1 = 3 - \frac{2}{6} = 2.6667$ M1; $x_2 = 2.6458$, so $\sqrt{7} \approx 2.646$ A1.
- 17 $f'(x) = 3x^2 - 2$ B1; $x_1 = 2.1, x_2 = 2.09457$ M1; root 2.095 A1.
- 18 $f'(x) = e^x - 3$ B1; $x_1 = 2.1873, x_2 = 1.9590$ M1 A1; $x_3 = 1.9064, x_4 = 1.9038$, root 1.9038 A1.



19 $f'(x) = \cos x - \frac{1}{2}$ B1; $x_1 = 1.9010$ M1 A1; $x_2 = 1.8955$, root 1.8955 A1.

20(a) $f(1.8) = -0.172$, $f(1.9) = 0.252$ M1; sign change, continuous A1.

20(b) $f'(x) = \frac{1}{x} + 2x$ B1; $x_1 = 1.8 - \frac{-0.1722}{4.1556}$ M1; $= 1.8414$ A1; $x_2 = 1.8411$ A1.

20(c) $f(1.84105) < 0$, $f(1.84115) > 0$ M1; sign change, so 1.8411 correct to 4 d.p. A1.

Section E – Failure

21 $f'(x) = 3x^2 - 3$, $f'(-1) = 0$ M1; division by zero / tangent is horizontal, so x_1 is undefined A1.

22 $f(x) = (x - 1.2)(x - 1.4)$ M1; roots 1.2 and 1.4 lie in (1, 2) A1; there are two roots, so the curve crosses the axis twice and the signs at the ends match B1.

23 $x_1 = 3.65$, $x_2 = 3.8645$, $x_3 = 4.1869$ M1 A1; moving away from 3 because $g'(3) = 1.2 > 1$ B1.

24 (a) $f'(x) = 3x^2 - 12x + 9$, $f'(3) = 0$, so division by zero M1 A1. (b) $x_1 = 2.9 - \frac{-2.971}{-0.57} = -2.31$ M1; 2.9 is close to a stationary point, so the tangent is nearly horizontal and sends x_1 far from the root A1.

25(a) $f(3) = -4$, $f(4) = 21$ M1; sign change, continuous A1.

25(b) $f'(x) = 3x^2 - 12$ B1; $x_1 = 3.2667$, $x_2 = 3.2337$ M1; $\alpha = 3.233$ A1.

25(c) $x_1 = 2.1 - \frac{-10.939}{1.23} = 10.99$ M1 A1; $x = 2$ is a stationary point, so the gradient near 2.1 is almost zero and the tangent meets the x -axis far away B1.

Roar-level challenge

26 $x_{n+1} = x_n - \frac{x_n^2 - a}{2x_n} = \frac{x_n^2 + a}{2x_n}$ M1; $= \frac{1}{2} \left(x_n + \frac{a}{x_n} \right)$ A1; $x_1 = 2.25$, $x_2 = 2.23611$, $x_3 = 2.23607$ M1; $\sqrt{5} = 2.2361$ A1.